

CPAN Custom k8s Implementation on the Cloud

CPAN

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Abstract

Phase 1 implementation includes the following steps:

- Custom k8s fully automated deployment on Azure:
 - Deploy Private Virtual Network with Private Virtual Subnets
 - Deploy Linux Virtual Machines (VM) (LB, k8s nodes)
 - Configure High Availability (HA) Load Balancer (LB)
 - Start Prime Master k8s node
 - Once Prime Master k8s node reach Ready state deploy rest of nodes (Masters, Workers)
- Custom k8s fully automated deployment on AWS (TODO)
- Custom k8s fully automated deployment on Google (TODO)

Chapter 1

Phase 1 Implementation

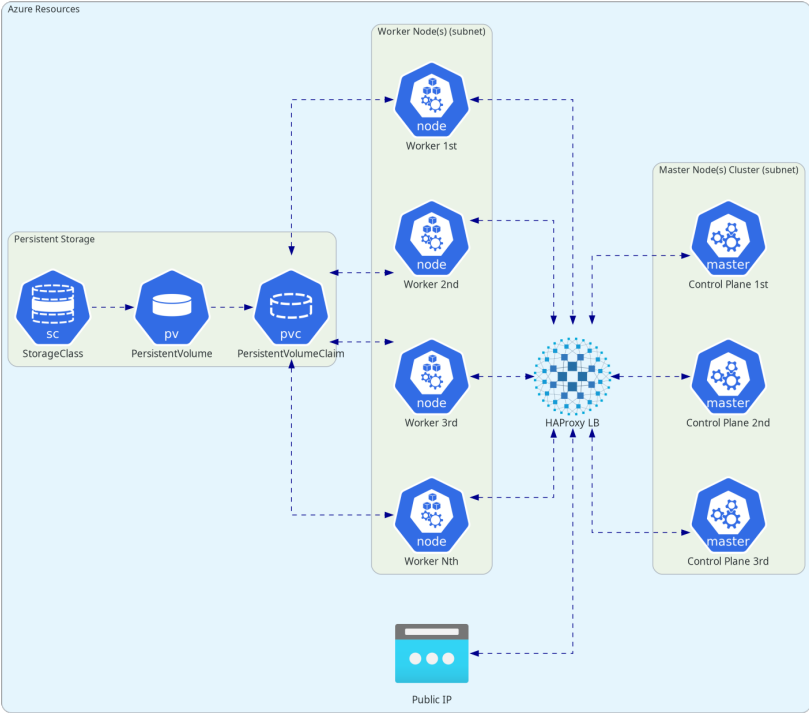
1.1 Two different possible scenarios:

- ✓ Local ETCD Infrastructure 1.1
- ✓ External ETCD Infrastructure 1.2

1.2 Fundamentals Benefits of a custom k8s cluster on the Cloud

- ✓ Reducing Cost: 💰
- ✓ Being able to customise the Operating System: 🐧
- ✓ Being able to customise the nodes: 🖥️

Figure 1.1: High Level Design Local ETCD



High Level Design - Azure custom K8s Infrastructure

Figure 1.2: High Level Design External ETCD

